

The Salsa20 Family of Stream Ciphers

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Abstract. Salsa20 is a family of 256-bit stream ciphers designed in 2005 and submitted to eSTREAM, the ECRYPT Stream Cipher Project. Salsa20 has progressed to the third round of eSTREAM without any changes. The 20-round stream cipher Salsa20/20 is consistently faster than AES and is recommended by the designer for typical cryptographic applications. The reduced-round ciphers Salsa20/12 and Salsa20/8 are among the fastest 256-bit stream ciphers available and are recommended for applications where speed is more important than confidence. The fastest known attacks use $\approx 2^{153}$ simple operations against Salsa20/7, $\approx 2^{249}$ simple operations against Salsa20/8, and $\approx 2^{255}$ simple operations against Salsa20/9, Salsa20/10, etc. In this paper, the Salsa20 designer presents Salsa20 and discusses the decisions made in the Salsa20 design.

1 Introduction

A sender and receiver share a short secret key. They use the secret key to encrypt a series of messages. A message could be short, just a few bytes, but it could be much longer, perhaps gigabytes. The series of messages could be short, just one message, but it could be much longer, perhaps billions of messages.

The sender and receiver encrypt messages using an **encryption function**: a function that produces the first ciphertext from the key and the first plaintext, that produces the second ciphertext from the key and the second plaintext, etc.

An encryption function has to be fast. Many senders have to encrypt large volumes of data in very little time using limited resources. Many receivers are faced with even larger volumes of data—not just the legitimate messages but also a flood of forgery attempts. A slow encryption function can satisfy some senders and receivers, but my focus is on encryption functions suitable for a wider range of applications.

An encryption function also has to be secure. Many users are facing, or at least think that they are facing, years of cryptanalytic computations by well-funded attackers equipped with millions of fast parallel processors. Some users

* Permanent ID of this document: `31364286077dcdff8e4509f9ff3139ad`. Date of this document: 2007.12.25. This work was supported by the National Science Foundation under grants CCR-9983950 and ITR-0716498, and by the Alfred P. Sloan Foundation.

Table 1. Salsa20 software speeds; measured by the official eSTREAM benchmarking framework; sorted by final column. “576” means single-core cycles/byte to encrypt a 576-byte packet; “long” means single-core cycles/byte to encrypt a long stream.

Arch	MHz	Machine	Salsa20 software	Cycles/byte					
				Salsa20/8 long	Salsa20/8 576	Salsa20/12 long	Salsa20/12 576	Salsa20/20 long	Salsa20/20 576
amd64	3000	Xeon 5160 (6f6)	amd64-xmm6	1.88	2.07	2.80	3.25	3.93	4.25
amd64	2137	Core 2 Duo (6f6)	amd64-xmm6	1.88	2.07	2.57	2.80	3.91	4.33
ppc32	533	PowerPC G4 7410	ppc-altivec	1.99	2.14	2.74	2.88	4.24	4.39
x86	2137	Core 2 Duo (6f6)	x86-xmm5	2.06	2.28	2.80	3.15	4.32	4.70
amd64	2000	Athlon 64 X2 (15,75,2)	amd64-3	3.47	3.65	4.86	5.04	7.64	7.84
ppc64	2000	PowerPC G5 970	ppc-altivec	3.28	3.48	4.83	4.87	7.82	8.04
amd64	2391	Opteron (f5a)	amd64-3	3.78	3.96	5.33	5.51	8.42	8.62
amd64	2192	Opteron (f58)	amd64-3	3.82	4.18	5.35	5.73	8.42	8.78
x86	2000	Athlon 64 X2 (15,75,2)	x86-1	4.50	4.78	6.27	6.55	9.80	10.07
x86	900	Athlon (622)	x86-athlon	4.61	4.84	6.44	6.65	10.04	10.24
ppc64	1452	POWER4	merged	6.83	7.00	8.35	8.51	11.29	11.47
hppa	1000	PA-RISC 8900	merged	5.82	5.97	7.68	7.85	11.39	11.56
amd64	3000	Pentium D (f64)	amd64-xmm6	5.38	5.87	7.19	7.84	10.69	11.73
x86	1300	Pentium M (695)	x86-xmm5	5.30	5.53	7.44	7.70	11.70	11.98
x86	3000	Xeon (f26)	x86-xmm5	5.30	5.86	7.41	8.21	11.64	12.55
x86	3200	Xeon (f25)	x86-xmm5	5.30	5.84	7.40	8.15	11.63	12.59
x86	2800	Xeon (f29)	x86-xmm5	5.33	5.95	7.44	8.20	11.67	12.65
x86	3000	Pentium 4 (f41)	x86-xmm5	5.76	6.92	8.12	9.33	11.84	13.40
x86	1400	Pentium III (6b1)	x86-mmx	6.37	6.79	8.88	9.29	13.88	14.29
sparc	1050	UltraSPARC IV	sparc	6.65	6.76	9.21	9.33	14.34	14.45
x86	3200	Pentium D (f47)	x86-athlon	7.13	7.66	9.90	10.31	15.29	15.94
ia64	1500	Itanium II	merged	8.49	8.87	12.42	12.62	18.07	18.27
ia64	1400	Itanium II	merged	8.28	8.65	12.56	12.76	18.21	18.40

are satisfied with lower levels of security, but again my focus is on encryption functions suitable for a wider range of applications.

There is a conflict between these desiderata. One can reasonably conjecture, for example, that every function that encrypts data in 0.5 Core-2 cycles/byte is breakable. One can also conjecture that *almost every* function that encrypts data in 5 Core-2 cycles/byte is breakable. On the other hand, several unbroken submissions to eSTREAM, the ECRYPT Stream Cipher Project, encrypt data in fewer than 5 Core-2 cycles/byte.

In particular, my 20-round stream cipher Salsa20/20 encrypts data in 3.93 Core-2 cycles/byte. (For comparison: Matsui and Nakajima recently reported 9.2 Core-2 cycles/byte for 10-round AES using a pre-expanded 128-bit key. See [18].) The fastest known attack against Salsa20/20 is a 256-bit brute-force search. I recommend Salsa20/20 for encryption in typical cryptographic applications.

Reduced-round ciphers in the Salsa20 family are attractive options for users who value speed more highly than confidence. The 12-round stream cipher Salsa20/12 encrypts data in 2.80 Core-2 cycles/byte; the fastest known attack against Salsa20/12 is a 256-bit brute-force search. The 8-round stream cipher